SAMPLE

Worlds Beyond Earth: The Sun

Our Star and the Source of All Life

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Free sample: Introduction & Chapter 1
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Introduction

The Sun is the sky's one certain star, the light every human has seen, the fixture so familiar that its strangeness takes deliberate effort to feel: a sphere of plasma wider than a hundred Earths, fusing six hundred million tons of hydrogen every second, holding every planet in a gravitational grip that reaches a hundred thousand times farther than the planet's own orbit, the sky's brightest object being also the nearest, the ordinary daylight being the most extraordinary fact in the audience's daily life.

This book tells the Sun's story in the campaign's usual way: the audience should keep the settled and the tentative apart, the wonder standing on measured footing. The chapters begin at the beginning, the star's size and the mind's failure to grasp it, and move outward through the structure, the core's fusion, the photosphere's grain, the atmosphere's layered heat, the activity's cycle, the spots and the flares and the great ejections, the wind that carries the Sun to every planet's door, the discoveries that made the Sun a star among stars, the spacecraft that have touched its outskirts, the survival question, the life question, the future's instruments, the book of facts, and the final approach, the imaginary journey the series always ends with, the whole account being the home star's biography written for the resident.

A note on the numbers, offered once for calibration: the Sun's measurements are the best in stellar astronomy, the star being the only one humanity has orbited, landed on by instruments, imaged in detail, and monitored continuously for centuries, and yet the traveler should hold the honest margins too, the interior being modeled rather than seen, the cycle being forecast poorly, the atmosphere's heating being the field's standing puzzle, the book carrying the field's own confidence levels chapter by chapter, the certain and the open kept apart, the way the campaign's books always keep them.

The introduction's surveyor should also meet the Sun's one role in the whole campaign, because the star recurs through the sibling volumes as the constant comparison. The planets' books borrowing the Sun as the landlord and the light source, the wind's books as the origin, the aurora books as the driver, the life books as the host, the Sun being the shelf's own hub, the hub taking its own spoke, this volume being the hub's own biography, the biography being the traveler's ground for the whole library, the ground being held first because the shelf stands on it.

The hub's visitor should hold the volume's own placement in the family's order too, because the Sun book is the shelf's cornerstone volume. The siblings' books being arranged by distance in the campaign's Worlds series, the inner planets' books borrowing the Sun's chapters, the outer volumes borrowing its light's faintness, the passenger who finishes the Sun volume holding the reference every other volume uses, the reference being the family's own spine, the spine being read first or last, the reading being the surveyor's own choice, the choice being the library's own freedom, the freedom being this paragraph's own point: the Sun holds no rank among the volumes, only the gravity, and the gravity being the reason the shelf orbits it.

The cornerstone's tourist should also meet the campaign's own Sun-first habit, because the other volumes' borrowings gets their list. The Mars volume borrowing the Sun's flux arithmetic, the Europa volume borrowing the habitable zone's edges, the aurora volumes borrowing the wind's mechanics, the space weather volumes borrowing the storm catalog, each sibling holding a promissory note drawn on this book, the note being paid here, the payment being the visitor's own collateral for the shelf, the shelf being the campaign's own system, the system being held together by the hub, the hub being the star, and the star being the reason the library exists at all.

The hub's bystander should close the placement note with the campaign's own honesty, because the shelf's spine has its humility: the Sun volume being the reference and not the crown, the crown belonging to the destinations, the Sun being the origin, the origin being read for grounding, the grounding being the passenger's own foundation course, the foundation being the point of reading the hub first, the first reading being recommended and the last reading being equally good, the good being the tourist's own shelf, the shelf being complete when the hub is held.

Chapter 01: Welcome to Our Star

Every introduction to the Sun should begin with the shock of scale, and this chapter is the shock: the Sun's size against the Earth's, the Sun's mass against the system's, and the Sun's membership in the Galaxy's stellar census, the familiar light being a monster by every measure except the Galaxy's.

The size claims the first statement, because the mind needs the numbers before the pictures. The Sun's diameter is about one and a half million kilometers, the Earth fitting across it a hundred and nine times, the volume holding

more than a million Earths, the wanderer should feel the conversion: if the Sun were a basketball, the Earth would be a pinhead two dozen meters away, the scale being the reason the Sun looks small, the distance being the reason the Sun feels ordinary, the two facts conspiring to hide the monster in plain sight, the daylight being the monster's touch, gentle at this range, the range being the solar system's own definition.

The mass merits the second statement, because the gravity is the mass. The Sun holds ninety-nine point eight percent of the solar system's mass, the planets and the moons and the asteroids sharing the remaining sliver, the explorer holding the proportion's meaning: the solar system is the Sun plus debris, the debris including the Earth, the hierarchy being absolute, every orbit being the Sun's grip, the grip reaching past the far comets, the Sun's light being the visible half and the Sun's gravity the invisible one, the system being a one-star show with the walk-ons in the wings.

The census completes the chapter, because the Sun is ordinary where it counts. The Galaxy holding hundreds of billions of stars, the Sun being a mid-sized, middle-aged, unremarkable member of the yellow-dwarf class, the sightseer should feel the double truth: the Sun is a monster against the planets and a mediocrity against the stars, the mediocrity being the luck, the quiet class being the long-lived one, the Sun's plainness being the Earth's tenure, the welcome chapter ending with the resident's own position, living on a speck in the grip of a middling star, the middling being the miracle, the ordinary star being the kind the universe does rarely and life needs always.

The census's student should feel the Sun's neighborhood too, because the mediocrity has company. The nearest stars being the small and the flare-prone class, the red dwarfs dominating the census, the Sun being larger than the great majority of the Galaxy's stars, the class holding the correction: the Sun is not the average star but the above-average one, the census's typical being the dim red speck, the Sun being the bright minority, the brightness being the rarity, the rarity being the Earth's own luck, the luck being auditable, the audit being the astrobiology's own arithmetic, the Sun being the census's generous outlier rather than its median resident.

The outlier's tour should take the Sun's class label once more with its own precision, because the class name is a habit. The yellow dwarf being the common shorthand, the star's true color being white, the yellow being the atmosphere's own tint at noon, the class being the main sequence's middle rung, the label being the astronomy's own filing, the pilot holding the naming's honesty: the Sun's whiteness being measurable from space, the yellowness being the eye's and the air's collaboration, the label being harmless and the truth being better, the star being a white star wearing the day's yellow, the naming chapter being the bystander's own correction to carry.

The color's stroller should take the noon's own physics too, because the yellow has its own daily proof. The air's scattering being the blue's own arithmetic, the removal of the blue leaving the yellow's own tint at noon and the red at dusk, the walker holding the color's whole account: the Sun's white light being the input, the sky's blue being the stolen band, the noon yellow and the sunset red being the receipts, the whole palette being the atmosphere's own edit of the star's true color, the edit being honest and the original being recoverable by altitude, the recovery being the astronaut's own privilege, the privilege being the sky's own proof of the star's whiteness.

The astronaut's audience should note the noon yellow's own companion fact too, because the sunrise and sunset belong to the same arithmetic. The low Sun's light traveling the long air path, the blue being scattered away entirely, the red surviving, the traveler holding the dusk's meaning: the star's apparent colors being one continuous edit, the noon gold and the evening crimson being the same white light at different tolls, the toll being the atmosphere's own tariff, the tariff being paid in color, the color account being the wanderer's own everyday physics, the physics being free with every clear dusk.

The star's distance also has its own card, because the light time is the first number to make the Sun real. The gap between the star and the planet is about a hundred and fifty million kilometers, a distance light itself, the fastest thing there is, needs more than eight minutes to cross, so the daylight warming any given moment left the star eight minutes before it arrived, and the Sun anyone ever sees is the Sun of eight minutes ago, the only past the eye can watch directly. The number is the whole book's first calibration: every fact in the chapters ahead is news, and even the light has a delay, the star being close enough to feel and far enough to give the wonder its measure.